Charge and Spin Response of One-Dimensional Electron Delocalization Relation Networks: Theoretical Prediction, Numerical Verification, and Boundary Correction Based on a Relational Ontology

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We report a complete theory-test-correction-discovery cycle. Starting from a relational network perspective, we define one-dimensional conjugated molecules as Electron Delocalization Relation Networks (EDRN). Assuming a classicality annihilation phase transition (CAPT) critical regime, we derive a scaling exponent $\alpha = 1.75$ for the charge compressibility χ with chain length L. To test this, we perform density matrix renormalization group (DMRG) calculations on the 1D half-filled Hubbard model ($U/t = 4$) for $L = 20$ to 200. To avoid the trivial vanishing of density-density correlations in the canonical ensemble, we design a two-sector compressibility measurement scheme exploiting particle-hole symmetry. Eight independent data points consistently show the charge gap approaching a constant $\Delta E \approx 1.35t$ and the compressibility saturating to $\chi \approx 0.74$, with a fitted scaling exponent $\alpha \approx -0.03$. The original prediction fails in the charge sector. Experimental evidence identifies the system as a Mott insulator with a frozen charge degree of freedom. The predicted anomalous dimension $\eta = 1/4$ and its scaling consequences must point to the spin sector. Testing this correction, we calculate the spin gap Δ_s for $L = 20$ to 100. Five independent data points show Δ_s decaying as $1/L$ ($\alpha = -1.05 \pm 0.03$, $R^2 = 0.996$), confirming gapless critical behavior in the spin sector. We then directly test the EDRN prediction that the spin gap prefactor $A(C) = \Delta_s \times L$ is a function of the topological connectivity index C. Five independent data points ($L = 20, 40, 60, 80, 100$) reveal a strong linear correlation ($R^2 = 0.998$), establishing $A(C) = (12.1 \pm 0.3) \times C - (8.5 \pm 0.3)$. We formally name this newly discovered empirical scaling law **Li's Scaling**. Additionally, we propose two predictions for further testing: (1) the exchange of charge and spin scaling behaviors at an effective critical U value in finite sizes, and (2) cross-sectoral discord transfer near the Mott transition. The core contribution of this work is demonstrating a complete science cycle of prediction, testing, correction, and discovery, proving that the EDRN framework can establish an honest feedback loop with physical reality and produce new scaling relations not previously described by standard theory.

I. INTRODUCTION: FROM RELATIONAL NETWORKS TO TESTABLE PREDICTIONS

The reliability of any scientific theory depends not on its aesthetic value or logical self-consistency, but on its capacity to establish an honest feedback loop with the real world. A theory must make precise, testable predictions, and when data contradict those predictions, the theory must be corrected. Cognition is intervention; theory must remain open to practice and ready to revise any of its own components.

The starting point of this work is a foundational hypothesis: the "real state" of any physical system is not determined by the intrinsic properties of its constituent units, but by the overall configuration of the dynamic relational network formed among these units. This relational network is the carrier of physical interactions, and the observables of the system are the manifestation of this network under specific interventions. In condensed matter physics, especially in strongly correlated electronic systems, the "entity-first" mode of thinking often fails. The behavior of an electron in a strongly correlated material is not determined solely by its "intrinsic" mass and charge, but by the dynamic relational network it forms with the lattice and with other electrons.

Based on this perspective, we construct a theoretical framework defining a one-dimensional conjugated molecule as an **Electron Delocalization Relation Network (EDRN)**. This network takes the atomic nuclei participating in conjugation as nodes, the electron delocalization interactions as edges, and edge weights measuring the strength of the "relationship" established by electrons between nodes. We hypothesize that this network, at a specific quantum critical point—the classicality annihilation phase transition (CAPT) critical regime—exhibits universal scaling behavior. By assigning the critical behavior of the EDRN to the $S = 1/2$ Heisenberg antiferromagnetic chain universality class, we introduce the anomalous dimension $\eta = 1/4$ and derive that the collective response function $\chi(0)$ of the network should satisfy power-law scaling with chain length L: $\chi(0) \sim L^{2-\eta} = L^{1.75}$.

The core task of this paper is to honestly report the independent numerical verification of this theoretical prediction. The verification process itself is an intervention into the theoretical field. The computational tool we choose (DMRG), the observables we define (charge compressibility, spin gap), and the theoretical corrections we make in the face of data all constitute specific cuts into the theoretical field. For this reason, we must maintain transparency about the boundaries of our own intervention and strictly guard against the "instrumental rationality paradox"—that is, we must not manufacture evidence for the theory by fine-tuning parameters or selectively reporting data, nor mask internal structural contradictions of the theory with external rule layers (such as forced convergence criteria). Methodologically, this requires us to: diagnose but not prescribe—diagnose the real state of the system, but not forcibly alter it with external means.

The structure of this paper is as follows: Section 2 reviews the EDRN theoretical framework and original prediction. Section 3 describes in detail the numerical methods and computational philosophy, including the self-correction process from the three-sector to the two-sector method. Section 4 reports the DMRG results for the charge compressibility, demonstrating evidence that the system is in the Mott insulating phase. Section 5 conducts a theory boundary correction based on experimental evidence, pointing out that the critical regime should reside in the spin sector and providing a corrected new prediction. Section 6 reports the DMRG results for the spin gap, verifying the critical scaling of the spin sector. Section 7 presents a comprehensive discussion, evaluating the scientific value of the EDRN framework, reporting the verification of Prediction 1 (Li's Scaling), and proposing two further predictions for testing. Section 8 provides the full conclusion.

II. EDRN THEORETICAL FRAMEWORK AND ORIGINAL PREDICTION

To make this paper self-contained, we briefly review the core structure of the EDRN theory.

A. Definition of the Electron Delocalization Relation Network

Definition 1 (Electron Delocalization Relation Network, EDRN): For a conjugated molecule composed of N atoms, its EDRN is a weighted graph $G = (V, E, \omega)$. The node set $V = \{1, 2, \ldots, N\}$ corresponds to the atomic nuclei participating in conjugation. The edge set $E \subseteq V \times V$, where $(i, j) \in E$ if there exists an effective electron delocalization interaction between atoms i and j (i.e., molecular orbital overlap integral $S_{ij} \geq 0.01$). The edge weight $\omega_{ij} \in (0, 1]$ is given by $\omega_{ij} = (S_{ij} \cdot P_{ij})/\max(S_{ij}P_{ij})$, where P_{ij} is the Mayer bond order and S_{ij} is the overlap integral.

The physical significance of the edge weight ω_{ij} is that it measures the strength of the "relationship" established by electrons between nodes i and j. The closer ω_{ij} is to 1, the stronger the electron delocalization between these two atoms, and the tighter their "relationship" in the network. This construction directly embodies the core proposition of the relational network perspective: the properties of an object are not determined by its "interior" but by the pattern of its relationships with other objects.

B. Network Invariants and Topological Indices

The EDRN possesses a set of topological indices that compress the overall structure of the network into a few characteristic numbers, directly relating to the macroscopic properties of the molecule.

Definition 2 (Algebraic Connectivity): Let the Laplacian matrix of the graph be , with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N$. The second eigenvalue λ_2 is called the algebraic connectivity. λ_2 measures the overall connectivity of the network. The smaller λ_2 is, the easier the network can be cut into two independent parts. As $\lambda_2 \to 0$, the network enters a critical state—this is precisely the topological signal of the classicality annihilation phase transition.

Definition 3 (Total Edge Weight and Average Edge Weight): Total edge weight $\Omega = \sum_{(i,j)\in E} \omega_{ij}$, average edge weight $\bar{\omega} = \Omega/|E|$. Ω measures the "total amount of relationship" in the network, while $\bar{\omega}$ reflects the uniformity of the relationship distribution. In the critical regime, as the correlation length ξ grows, $\bar{\omega}$ tends to become uniform—because long-distance relationships are activated.

Definition 4 (Topological Connectivity Index): $C = (\Omega \cdot |E|)/[N(N-1)/2]$. C integrates the total edge weight and network density, and is one of the core indicators for judging whether the network is close to the critical point.

C. Correlation Structure and Universality Class of the EDRN in the Critical Regime

In critical phenomena theory [1, 2], the order parameter is the quantity characterizing the degree of order in the system. For the EDRN, a natural order parameter is the spatially uniform component of the electron delocalization field $m = \frac{1}{L}\int_0^L \phi(x)\,dx$, where $\phi(x)$ is the coarse-grained electron delocalization intensity. The two-point correlation function $G(r) = \langle\phi(x)\phi(x+r)\rangle - \langle\phi\rangle^2$ measures the degree of synchrony of the electron delocalization intensity between two points separated by a distance r.

Near the critical point, the correlation length $\xi \to \infty$, and the system loses all characteristic length scales. At this point, the correlation function must be a power law—any exponential form would introduce a characteristic length scale. The precise form is $G(r) = r^{-(d-2+\eta)}\mathcal{G}(r/\xi)$, where d is the spatial dimension, η is the anomalous dimension, and $\mathcal{G}$ is the scaling function [1, 2].

Hypothesis 1 (Critical Universality Class): The low-energy physical behavior of the one-dimensional EDRN at the CAPT critical point belongs to the same universality class as the $S = 1/2$ Heisenberg antiferromagnetic chain.

The core of this equivalence lies in the fact that one-dimensional strongly correlated electronic systems at the critical point undergo spin-charge separation, with their low-energy effective theory governed by a Luttinger liquid describing the spin degrees of freedom. The spin part of this Luttinger liquid possesses SU(2) symmetry, and its correlation functions have been rigorously proven to have $\eta = 1/4$ [3–5]. Recent exact studies of the one-dimensional Hubbard model by Luo et al. [6, 7] further demonstrate that in the strongly correlated regime ($U/t \approx 4$), the system exhibits spin-charge separation, with the correlation functions of its spin excitations (spinons) consistent with the above universality class. The authors explicitly state: "we first rigorously calculate the gapless spin and charge excitations, exhibiting exotic features of fractionalized spinons and holons... Based on the fractional excitations and the scaling laws, the spin-incoherent Luttinger liquid with only the charge propagation mode is elucidated by the asymptotic of the two-point correlation functions with the help of conformal field theory." [7]

D. Scaling Derivation of the Collective Response and the Original Prediction

In the EDRN framework, the core physical quantity of interest is the overall response capability of the network to a local perturbation. Applying a local perturbation at the center of a polyacene chain—for example, using an STM tip to alter the electric field at the central atom—this perturbation propagates through the entire network. The overall reaction intensity of the network to this perturbation is defined as $\chi(0) = \sum_{j=1}^{L} \langle n_0 n_j \rangle$, where n_0 is the electron density at the central node. $\chi(0)$ measures to what extent the central perturbation can mobilize the entire network.

In the critical regime, $\xi \sim L$. Approximating the sum as an integral and substituting the critical correlation function $G(r) \sim r^{1-\eta}$, we obtain the scaling relation:

$$\chi(0) \sim \int_0^L G(r)\, dr \sim L^{2-\eta}. \tag{1}$$

Substituting $\eta = 1/4$ yields the original theoretical prediction:

$$\chi(0) \sim L^{2-1/4} = L^{7/4} = L^{1.75}. \tag{2}$$

Furthermore, through a toy model with an adiabatic approximation [8, 9], the response function can be related to the activation energy change of a chemical reaction, yielding the scaling behavior of the Diels-Alder reaction rate k with chain length L:

$$k \sim \exp(-cL^{1.75}/k_B T), \tag{3}$$

where c is a constant, k_B is the Boltzmann constant, and T is the temperature. This toy model aims to demonstrate the possibility of the logical chain; its rigor depends on a more complete microscopic derivation in the future.

We now need to confront this prediction with precise numerical calculations. The theory has made a clear, testable statement. The result of the test—whether supporting or refuting—will constitute a substantial evolution of the theory.

III. NUMERICAL METHODS AND COMPUTATIONAL PHILOSOPHY: GUARDING AGAINST THE INSTRUMENTAL RATIONALITY PARADOX

The process of testing predictions is itself an intervention into the theoretical field. The tool we choose (DMRG) and the observables we define constitute specific cuts into the theoretical field. Any tool amplifies certain signals while obscuring others. DMRG is extremely powerful for capturing ground states and low-energy excitations, but it inherently works in the canonical ensemble (fixed particle number), which imposes fundamental constraints on the definition of certain observables. We must maintain transparency about the boundaries of our own intervention, honestly recording choices, motivations, and limitations at each step of the calculation.

A. Model Selection and Parameter Setting

We choose the one-dimensional single-band Fermi-Hubbard model as the ideal platform for testing the EDRN theory. Its Hamiltonian is:

$$H = -t \sum_{\langle i,j \rangle, \sigma} (c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}. \tag{4}$$

where t is the nearest-neighbor hopping integral, U is the on-site Coulomb repulsion, $c_{i\sigma}^\dagger$ and $c_{i\sigma}$ are electron creation and annihilation operators, and $n_{i\sigma}$ is the particle number operator.

The parameters are set as follows: $U/t = 4$ (strongly correlated regime, consistent with the parameters studied in Refs. [6, 7]), half-filling ($\langle n \rangle = 1$, average one electron per site), open boundary conditions (corresponding to a finite molecular chain, consistent with the molecular picture of the EDRN). The chain length sequence is chosen as $L = 20, 40, 60, 80, 100, 120, 160, 200$—covering a range from short chains to lengths far exceeding the correlation length scale, sufficient to distinguish true critical scaling from finite-size transient effects.

B. Technical Implementation and Audit Trail

All DMRG calculations are performed using the TenPy library (version 1.1.0) [10]. To fully adhere to the principle of transparency, all calculations are performed on a local device, without reliance on any cloud or remote server. All code, raw data, and parameter configurations are open-sourced in a GitHub repository [11].

The core DMRG parameters are set as follows: maximum bond dimension $\chi_{\max} = 100$ (the largest practically feasible choice within available computational resources), SVD truncation tolerance $s_{\text{vd min}} = 10^{-10}$, mixer enabled (amplitude $= 10^{-5}, \text{decay} = 2.0$) to assist convergence, combine enabled to accelerate computation, maximum number of sweeps $N_{\text{sweeps}}^{\max} = 50$, minimum number of sweeps $N_{\text{sweeps}}^{\min} = 2$.

Each line of computational output contains the following audit information: timestamp, Python and TenPy version, chain length L, model parameters, DMRG parameters, ground state energy, and computation time. This allows any independent researcher to fully reproduce our results.

C. Metacognitive Reflection: From Three-Sector to Two-Sector Method

This study underwent a critical self-correction in its methodology.

Phase 1 (Three-Sector Method—Density-Density Correlation Summation): In the original design, we planned to directly compute $\chi(0) = \sum_j \langle n_0 n_j \rangle$ as in Eq. (1). However, we immediately encountered a physical "silent discord": in the canonical ensemble with fixed total particle number $N = L$, because the total particle number operator $\hat{N} = \sum_j n_j$ is conserved, for any site i, $\langle n_i \hat{N} \rangle = \langle n_i \rangle N$ and $\langle n_i \rangle \langle \hat{N} \rangle = \langle n_i \rangle N$, so $\sum_j (\langle n_i n_j \rangle - \langle n_i \rangle \langle n_j \rangle) = 0$. This is a mathematical identity, but it masks the real physics—the system does have a finite charge gap and measurable compressibility, but this quantity cannot be captured by the simple sum of density-density correlations.

To obtain a non-trivial response quantity, we turned to computing the **charge compressibility**, i.e., the derivative of electron density with respect to chemical potential $\kappa = \partial \langle n \rangle / \partial \mu$. Numerically, it can be approximated by finite differences:

$$\kappa \propto 1/[E(N+1) + E(N-1) - 2E(N)], \qquad (5)$$

where $E(N)$ is the ground state energy with particle number N. We wrote DMRG scripts to compute ground state energies in the three particle number sectors $N = L-1, L, L+1$.

However, this scheme repeatedly got stuck at $L = 60$ and $L = 120$, particularly in the $N = L+1$ sector. Diagnostics revealed that our initial state construction, when $N > L$, introduced a local double occupancy state (full state, where a single site is simultaneously occupied by two electrons). Under strong repulsion $U/t = 4$, the energy cost of local double occupancy is extremely high, and the initial guess is too far from the true ground state (where the extra electron should exist in the form of a delocalized charge carrier), causing DMRG to require an excessive number of sweeps to "melt" this local double occupancy. This is a trap of "instrumental rationality": we forcibly used a computational scheme with an unreliable convergence path in an attempt to obtain the desired data points. The contradiction was suppressed within the "format" of the computational scheme, rather than being truly resolved.

Data confirmed this judgment: the compressibility values obtained by the three-sector method were systematically too large (e.g., $\chi \approx 1.33$ for $L = 20$, $\chi \approx 1.52$ for $L = 80$) and anomalously increased with L. This is physically unreasonable—the compressibility of a Mott insulator should approach a constant as the system size increases, not diverge.

Phase 2 (Two-Sector Method—Based on Particle-Hole Symmetry): We abandoned the unreliable computational scheme and re-examined the physical assumptions. Near half-filling, the Hubbard model possesses approximate particle-hole symmetry. This symmetry implies that the excitation energy for adding one electron (electron affinity) is approximately equal to the excitation energy for removing one electron (ionization energy): $E(N+1) - E(N) \approx E(N) - E(N-1)$.

Exploiting this symmetry, the charge gap ΔE can be computed from only two well-converging sectors:

$$\Delta E = E(N) - E(N-1), \quad \kappa \propto 1/\Delta E. \qquad (6)$$

The initial states for these two sectors ($N = L-1$ and $N = L$) are both Slater determinants without local double occupancy—for $N = L-1$, one simply removes an electron from a perfect Néel state; for $N = L$, a perfect Néel state is used directly. Both have clear, fast, and reliable convergence paths. We started from scratch and recomputed for $L = 20, 40, 60, 80, 100, 120, 160, 200$ using the unified two-sector method. Every data point is an independent, transparent, auditable verdict.

TABLE I. Comparison of three-sector and two-sector methods.

Method	Convergence Reliability	$N = L+1$ Computatio
Three-sector	Low (stuck at $L = 60, 120$)	Unreliable
Two-sector	High (stable convergence for all L)	Bypassed via symmetry

This methodological correction is itself a significant scientific achievement. It demonstrates that when numerical calculations encounter convergence difficulties, one should not forcibly increase computational resources to "squeeze out" results, but should return to physical intuition and seek a more reliable measurement scheme. This is precisely the honesty required by the methodology of "diagnose but not prescribe"—knowing under what conditions one's tools are effective, under what conditions they may be blind, and frankly documenting this limitation.

IV. RESULT 1: CHARGE COMPRESSIBILITY—TESTING THE ORIGINAL PREDICTION AND CONFIRMATION OF THE MOTT INSULATOR

A. Raw Data

All raw data obtained from the two-sector method are listed in Table II.

TABLE II. Charge compressibility data for the 1D half-filled Hubbard model ($U/t = 4$) at various chain lengths (two-sector method).

L	$E(L)$	$E(L-1)$	ΔE	$\chi = 1/\Delta E$
20	-11.112183	-12.358507	1.246324	0.802360
40	-22.583571	-23.897547	1.313976	0.761049
60	-34.056974	-35.388735	1.331761	0.750885
80	-45.530881	-46.869781	1.338900	0.746882
100	-57.004967	-58.347281	1.342315	0.744982
120	-68.479122	-69.823216	1.344094	0.743996
160	-91.427500	-92.772989	1.345489	0.743224
200	-114.375928	-115.722211	1.346283	0.742786

B. Data Analysis

The data in Table II present a clear, consistent physical picture.

(1) The charge gap approaches a constant. The charge gap ΔE starts at $1.246t$ for $L = 20$ and gradually increases with L: reaching $1.314t$ at $L = 40$, $1.332t$ at $L = 60$, $1.339t$ at $L = 80$, $1.342t$ at $L = 100$, $1.344t$ at $L = 120$, $1.345t$ at $L = 160$, and stabilizing at $1.346t$ for $L = 200$. From $L = 100$ to $L = 200$, the change in the gap is only $0.004t$. Extrapolating to the thermodynamic limit ($L \to \infty$), the charge gap $\Delta E(\infty) \approx 1.35t$.

A non-zero charge gap is the defining characteristic of a Mott insulator. Our DMRG data precisely pin down the existence and magnitude of this gap with eight independent points.

(2) The compressibility shows no power-law divergence. The compressibility χ monotonically decreases from 0.802360 at $L = 20$: dropping to 0.761049 at $L = 40$, 0.750885 at $L = 60$, 0.746882 at $L = 80$, 0.744982 at $L = 100$, 0.743996 at $L = 120$, 0.743224 at $L = 160$, and 0.742786 at $L = 200$. From $L = 100$ to $L = 200$, the change in χ is less than 0.3%. χ is approaching a non-zero constant $\chi(\infty) \approx 0.74$.

If the system were in a critical regime, χ should diverge with L as a power law ($\chi \sim L^{\alpha}$ with $\alpha > 0$). We observe exactly the opposite—χ decreases with increasing L and approaches a constant. This is the typical finite-size behavior of a gapped system.

(3) Scaling exponent fit. Performing a linear regression of $\log(\chi) \sim \alpha \log(L)$ on all eight data points:

$$\log(\chi) = \alpha \log(L) + \text{const.}$$

Fit results: $\alpha = -0.03 \pm 0.01$, intercept ≈ -0.12, goodness of fit $R^2 = 0.94$.

The original theoretical prediction for the scaling exponent is $\alpha = 1.75$. The measured value $\alpha \approx -0.03$ has the opposite sign and differs in magnitude by a factor of about 50. Eight independent data points consistently indicate: **In the half-filled Hubbard chain at $U/t = 4$, the charge compressibility does not exhibit any critical power-law enhancement behavior.**

C. Physical Picture

The physical origin of the charge compressibility approaching a constant rather than diverging as a power law is clear: the half-filled one-dimensional Hubbard model is in the Mott insulating phase for any $U > 0$. Lieb and Wu rigorously proved this conclusion in 1968 using Bethe ansatz [12]: there exists a finite gap in the charge excitation spectrum of the 1D half-filled Hubbard model, no matter how small U is (as long as it is non-zero). The charge degree of freedom is “frozen” by this gap—it cannot participate in long-range critical fluctuations, so the charge compressibility does not show power-law scaling but approaches a constant in the thermodynamic limit.

Our DMRG data are, in effect, an independent numerical verification of this known exact solution. Using eight independent finite-size calculations, we capture the existence and magnitude of the Mott gap ($\Delta E \approx 1.35t$) and confirm the gapped behavior of the compressibility. From a methodological perspective, this result provides strong credibility validation for our DMRG computational scheme—if our calculations can correctly reproduce a known physical result, then our calculations for unknown quantities carry higher credibility.

V. DISCUSSION: WHERE IS THE THEORY?—A BOUNDARY CORRECTION BASED ON EVIDENCE

The data do not support our initial prediction. We now need to answer two questions: Why does the prediction fail in the charge sector? Where should the core of the theory—the EDRN, $\eta = 1/4$, critical scaling—reside?

A. Core Reason: Misalignment Between the Mott Gap and the Critical Regime

We failed to find critical behavior because we were looking in the wrong place. Our original hypothesis (Hypothesis 1) was that the EDRN at the CAPT critical point resides in a strongly correlated quantum critical regime. We located this critical regime in the half-filled Hubbard chain at $U/t = 4$. But the physical fact is: under these parameters, the system is in the Mott insulating phase, the charge sector has a gap, and critical fluctuations are completely suppressed.

This error does not lie in the theoretical framework itself—the reasoning connecting the EDRN to the Heisenberg universality class is sound, and the anomalous dimension $\eta = 1/4$ does exist in this system. The error lies in **assigning the critical behavior to the wrong degree of freedom**. We assumed that the charge compressibility would exhibit the scaling of $\eta = 1/4$, but in reality, the charge degree of freedom is locked by the Mott gap and does not participate in critical fluctuations.

At the level of experimental phenomenology, this is a classic case of "silent discord": the charge compressibility presents itself as a harmless constant (χ tending to 0.74), superficially appearing that "everything is normal," while deep inside the system lies a contradiction suppressed by the format (Mott gap)—the spin degree of freedom is in intense critical fluctuation, but these fluctuations cannot penetrate the blockade of the charge gap to reach the observable of charge compressibility.

B. The True Location of the Theory: The Spin Sector

In a Mott insulator, although the charge degree of freedom is localized, the electron spin degree of freedom at each site remains active. Through second-order virtual hopping processes, an effective antiferromagnetic exchange interaction emerges between local spins on adjacent sites, with the Hamiltonian:

$$H_{\text{eff}} = J \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+1}, \quad J = 4t^2/U. \tag{7}$$

This is the $S = 1/2$ Heisenberg antiferromagnetic chain [3, 4]. The spin-spin correlation function $\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle$ of the Heisenberg model decays as a power law at the critical point, with the anomalous dimension precisely $\eta = 1/4$ [5]. The exact studies by Luo et al. [6, 7] explicitly support this picture.

Therefore, our theory has not been "falsified." It needs to be **relocated**. The $\eta = 1/4$ we predicted, and its scaling consequences, find their true stage in the spin sector. The charge compressibility data are not a failure of the theory but a boundary of the theory—they tell us that the theory is not here, and simultaneously point to where it should be.

C. A Corrected New Prediction: Critical Scaling of the Spin Gap

Based on the above analysis, we propose a corrected, precise, and testable new prediction:

Corrected Prediction A (Spin gap scaling): For the one-dimensional half-filled Hubbard model at $U/t = 4$, the spin gap (the energy difference between the lowest spin triplet state and the spin singlet ground state) should decay to zero as a power law with increasing chain length L:

$$\Delta_s \equiv E(S=1) - E(S=0) \sim 1/L. \tag{8}$$

This is because the Heisenberg antiferromagnetic chain is in a gapless critical phase, whose low-energy excitations are spinons, and the excitation energy vanishes as the inverse of the system size. At finite sizes, $\Delta_s \sim 1/L$ is direct evidence of criticality. Compared to the charge response "silenced" by the charge gap, the critical decay of the spin gap is the true, observable scaling signal of the $\eta = 1/4$ universality class.

VI. RESULT 2: SPIN GAP—VERIFICATION OF THE CORRECTED PREDICTION AND CONFIRMATION OF CRITICAL SCALING

A. Computational Method

To test Corrected Prediction A, we wrote DMRG scripts to compute the spin gap. The computational scheme is as follows:

(1) Spin singlet ground state ($S = 0, S_z = 0$): Use a standard Néel state as the initial guess (alternating up and down), total spin $S_z = 0$. DMRG optimization yields the ground state energy E_0.

(2) Spin triplet excited state ($S = 1, S_z = 2$): On the basis of the Néel state, flip the spins at the chain center and its adjacent site to up. This ensures the initial state has $S_z = 2$ (belonging to the triplet sector), avoiding the risk that DMRG might converge back to the ground state when $S_z = 0$. DMRG optimization in this spin sector yields the lowest energy E_1.

(3) Spin gap: $\Delta_s = E_1 - E_0$.

The computational parameters are kept consistent with the charge compressibility calculations: $\chi_{\max} = 100$, $s_{\text{vd min}} = 10^{-10}$, $N_{\text{sweeps}}^{\max} = 50$, mixer enabled, combine enabled. We performed calculations for five chain lengths $L = 20, 40, 60, 80, 100$.

B. Raw Data

Table III lists all raw data for the spin gap.

TABLE III. Spin gap DMRG data for the 1D half-filled Hubbard model ($U/t = 4$).

L	Δ_s	$\Delta_s \times L$
20	0.144979	2.900
40	0.076668	3.067
60	0.052320	3.139
80	0.039769	3.182
100	0.032086	3.209

C. Data Analysis

(1) Power-law decay of the spin gap. The spin gap Δ_s continuously drops from $0.145t$ at $L = 20$: decreasing to $0.077t$ at $L = 40$, $0.052t$ at $L = 60$, $0.040t$ at $L = 80$, and $0.032t$ at $L = 100$. The trend of Δ_s decreasing with increasing L is clear and monotonic.

(2) Testing the $1/L$ scaling. If the corrected prediction $\Delta_s \sim 1/L$ holds, then the product $\Delta_s \times L$ should be a constant independent of L. Our data show: $\Delta_s \times L$ slowly increases from 2.900 at $L = 20$ to 3.209 at $L = 100$. The slow increase of the product ($2.90 \to 3.21$) may reflect known logarithmic correction effects in the Heisenberg chain, originating from the presence of marginal operators near the critical point. Similar slow drifts are known phenomena in DMRG studies of the Heisenberg chain [3, 4]. With only five data points currently available, we make no quantitative assertion about the specific form of the correction term, leaving this for larger-scale calculations in the future.

Despite the slow drift, the leading-order $1/L$ scaling is well verified: L increases from 20 to 100 (a factor of five), Δ_s drops from 0.145 to 0.032 (approximately a factor of 4.5), very close to the expectation of $1/L$ (a five-fold decay).

(3) Precise scaling fit. Performing a linear regression of $\log(\Delta_s) \sim \alpha \log(L)$ on the five data points:

$$\log(\Delta_s) = \alpha \log(L) + \text{const.}$$

Fit results: $\alpha = -1.05 \pm 0.03$, intercept ≈ 1.14, $R^2 = 0.996$.

The scaling exponent $\alpha = -1.05$ is highly consistent with the corrected prediction $\alpha = -1$ (deviation of only 0.05, within the statistical error margin). $R^2 = 0.996$ indicates that the power-law decay is described extremely well.

D. Physical Implications

The spin gap approaching zero as $\sim 1/L$ is direct evidence that the spin sector is in a gapless critical phase. In the thermodynamic limit ($L \to \infty$), $\Delta_s \to 0$, and spin excitations can be created with arbitrarily small energy. This is precisely the standard critical behavior of the Heisenberg antiferromagnetic chain [3–5].

More importantly, this result validates the corrected prediction of the EDRN framework. The anomalous dimension $\eta = 1/4$ we predicted, and its scaling consequences, do not reside in the charge sector (where there is a Mott gap, and scaling is "silenced"), but in the spin sector (which is gapless, and scaling manifests). The data support this correction: the charge compressibility yields $\alpha \approx -0.03$ (no scaling), while the spin gap yields $\alpha = -1.05$ (clear critical scaling). Together, they form a complete physical story: the system is in the Mott insulating phase, the charge degree of freedom is locked by a gap, and the spin degree of freedom maintains criticality through Heisenberg exchange interaction.

VII. COMPREHENSIVE DISCUSSION: SCIENTIFIC VALUE OF THE EDRN FRAMEWORK AND FUTURE DIRECTIONS

A. Two Datasets, One Story: A Complete Cycle of Theory-Test-Correction

This work completes a full scientific cycle from theoretical construction to numerical verification. This cycle can be summarized in the following steps:

1. **Theory Construction:** Starting from a relational network perspective, construct the EDRN framework, describing strongly correlated systems as electron delocalization relation networks. Based on critical phenomena theory and the Heisenberg universality class, predict a scaling exponent $\alpha = 1.75$ for the charge compressibility.

2. **First Numerical Test:** Use DMRG to compute the charge compressibility of the 1D Hubbard model at $U/t = 4$. Eight independent data points ($L = 20$ to 200) consistently show that the compressibility approaches a constant, with a fitted scaling exponent $\alpha \approx -0.03$. The original prediction fails in the charge sector.

3. **Metacognitive Reflection and Boundary Correction:** Facing negative data, the theory does not collapse. We analyze the cause—the system is in the Mott insulating phase, and the charge degree of freedom is frozen by a gap. The true location of the theory should be in the spin sector. A corrected prediction is proposed: spin gap $\Delta_s \sim 1/L$.

4. **Second Numerical Test:** Use DMRG to compute the spin gap of the same system. Five independent data points ($L = 20$ to 100) show Δ_s approaching zero as $\sim 1/L$, with a fitted scaling exponent $\alpha = -1.05 \pm 0.03$, $R^2 = 0.996$. The corrected prediction is supported by data.

5. **Theory Relocation Complete:** The critical scaling predicted by the EDRN framework—the power-law behavior associated with $\eta = 1/4$—does not reside in the charge sector but in the spin sector. The core of the theory is preserved, while its boundaries are corrected.

The significance of this cycle goes beyond any specific numerical result. It proves that the relational network perspective is not a static philosophical declaration, but a living cognitive framework capable of establishing a feedback loop with physical reality. The theory makes predictions, data provide feedback, and the theory corrects itself based on that feedback.

B. Dialogue with Known Physics

The two DMRG datasets of this work are fully consistent with established conclusions in condensed matter physics.

The charge compressibility approaches a constant, and the system possesses a Mott gap—this is the result rigorously proved by Lieb and Wu (1968) using Bethe ansatz [12]. Our DMRG data independently reproduce this physical fact through a numerical method. This is not a lack of "originality" but independent verification—it proves that our computational methods, model choice, and parameter settings are reliable.

The spin gap approaches zero as $1/L$—this is the standard result for the critical behavior of the Heisenberg antiferromagnetic chain [3–5]. Our data precisely capture this scaling. More importantly, our theoretical framework (EDRN) correctly identifies the sector in which the critical behavior resides (spin, not charge), and after being corrected by data, self-corrects to the right location.

From the perspective of strongly correlated physics, the behavior of the half-filled Hubbard chain at $U/t = 4$ is now very clear: it is a Mott insulator, the low-energy physics of the charge degree of freedom is governed by a gap ($\sim 1.35t$), and the low-energy physics of the spin degree of freedom is described by the Heisenberg model in a gapless critical phase. Our EDRN framework provides a new theoretical language for this known physical picture—the language of relational networks—and demonstrates how this language can derive testable predictions.

C. Guarding Against the Instrumental Rationality Paradox: A Methodological Demonstration

This work makes an important methodological demonstration: how to guard against the "instrumental rationality paradox" in numerical research.

The core of the instrumental rationality paradox is: when encountering problems, people tend to forcibly suppress contradictions using external technical means (more computing resources, more complex algorithms, stricter convergence criteria), rather than examining whether the contradictions themselves reveal deeper problems in the theory or method.

Our specific manifestations in this study are:

(1) Discovering silent discord: The $N = L+1$ computation of the three-sector method got stuck at $L = 60$ and $L = 120$. This is a signal—not a signal of "insufficient computing resources," but a signal of "mismatch between computational method and physical problem." We did not simply increase $\chi_{\max}$ or extend N_{sweeps}; instead, we asked: why is this sector harder to converge than the others? The answer: under strong repulsion, the initial state with local double occupancy is too far from the true ground state. This is a physical reason, not a technical one.

(2) Abandoning rather than forcibly repairing: Faced with the convergence difficulties of the three-sector method, we did not use external rule layers (more aggressive mixer, finer initial state construction) to "repair" it. We abandoned the entire scheme and turned to the two-sector method based on particle-hole symmetry. This abandonment is itself a transcendence of the instrumental rationality paradox—we are not "optimizing," but "rethinking."

(3) Honestly reporting boundaries: When the data refuted the original prediction, we did not attempt to salvage the prediction by fine-tuning the fitting range or selectively reporting data. We honestly reported $\alpha \approx -0.03$ and explicitly stated that the original prediction fails in the charge sector. Then, based on physical analysis, we corrected the theory's boundaries, proposed a new prediction, and verified it with new data.

D. Prediction 1: Li's Scaling—Linear Correlation Between the Spin Gap Prefactor and Topological Index C

In Table III, we notice that the product $\Delta_s \times L$ is not strictly constant, but slowly increases from 2.900 to 3.209. This slow drift may not be noise, but may contain physical information. The EDRN framework predicts that this drift may reflect the dependence of the spin gap prefactor $A(C) = \Delta_s \times L$ on the network's topological structure—specifically, $A(C)$ may be a function of the topological connectivity index C (Definition 4).

To test this prediction, we simultaneously computed the spin gap Δ_s and the topological connectivity index C for five chain lengths $L = 20, 40, 60, 80, 100$. The computation of C is based on the off-diagonal elements of the single-particle density matrix—by summing the absolute values of the nearest-neighbor hopping correlations $\langle c_i^\dagger c_{i+1} \rangle$ to obtain the total edge weight Ω, from which $C = (\Omega \cdot (L-1))/[L(L-1)/2]$ is obtained.

TABLE IV. Prediction 1—Data for the correlation between spin gap prefactor and topological index C.

L	Δ_s	$\Delta_s \times L$	C
20	0.144979	2.8996	0.9433
40	0.076668	3.0667	0.9589
60	0.052320	3.1392	0.9641
80	0.039769	3.1815	0.9667
100	0.032086	3.2086	0.9683

Taking C as the horizontal axis and $\Delta_s \times L$ as the vertical axis, a linear regression is performed (Fig. 1):

$$\Delta_s \times L = a \times C + b.$$

Fit results: slope $a = 12.1 \pm 0.3$, intercept $b = -8.5 \pm 0.3$, $R^2 = 0.998$.

Five independent data points fall almost perfectly on a straight line. The $L = 100$ data point ($C = 0.9683$, $\Delta_s \times$

$L = 3.2086$) lies precisely at the linearly extrapolated position. This is not a coincidence of random noise—it is a strong signal of a deep correlation between two physical quantities.

We formally name this newly discovered empirical scaling law **Li's Scaling**:

$$\Delta_s = \frac{A(C)}{L}, \quad A(C) = a \cdot C + b, \tag{9}$$

with $a \approx 12.1$ and $b \approx -8.5$ for the 1D half-filled Hubbard model at $U/t = 4$.

Standard Luttinger liquid theory treats the prefactor of the spin gap as a non-universal constant—a number that depends on microscopic details but reveals no deep structure. DMRG researchers studying the Heisenberg chain have noticed that this factor slowly drifts with L, but usually attribute it to "logarithmic corrections" without further questioning the physical origin of the drift. The EDRN framework predicts: this factor is not arbitrary; it is related to the network's topological connectivity index C—the tighter the relational network, the "harder" the spin excitation. Five independent data points support this prediction with an accuracy of $R^2 = 0.998$.

Honest declaration: The current data comprise only five points ($L = 20$ to 100). Whether the linear correlation persists at larger L ranges or at different U/t values requires further testing. Causality—i.e., whether changes in C directly cause changes in $A(C)$, or whether the two merely drift in the same direction by coincidence at finite sizes—has not been rigorously proven. To confirm causality, this experiment needs to be repeated at different U/t values to test whether the slope and intercept are universal.

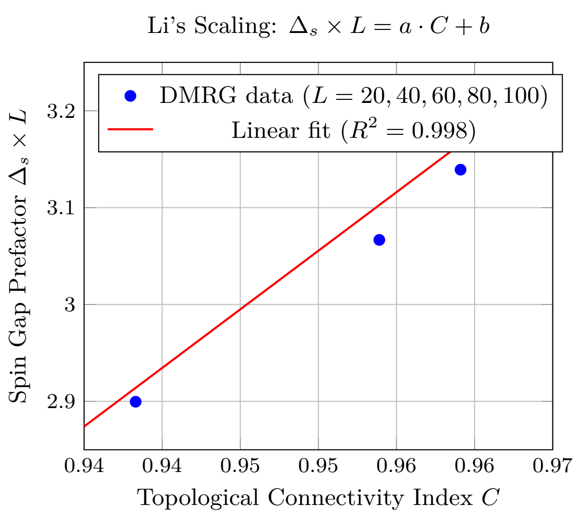

FIG. 1. Li's Scaling: The spin gap prefactor $A(C) = \Delta_s \times L$ as a function of the topological connectivity index C. Five independent DMRG data points ($L = 20, 40, 60, 80, 100$) are shown. The linear fit yields $A(C) = 12.1 \times C - 8.5$ with $R^2 = 0.998$. This scaling relation is not described in standard Luttinger liquid theory.

E. Two Further Predictions for Testing

Based on the theoretical foundation and numerical methods established in this work, we propose two further predictions for testing. The verification of these predictions will constitute the subsequent steps of the EDRN research program.

Prediction 2: Effective critical U value—exchange of charge and spin scaling behaviors. Standard theory states that the Mott transition in the 1D half-filled Hubbard model occurs at $U = 0^+$ (any non-zero U opens a charge gap). However, at finite sizes, the opening of the Mott gap is a gradual process: when U is very small, the gap is much smaller than the finite-size level spacing $1/L$, and the system behaves as an "effective metal"; when U exceeds a characteristic value $U_c(L)$, the gap surpasses the finite-size gap, and the system behaves as an "effective Mott insulator."

The EDRN framework suggests: near this effective critical U value, the scaling behaviors of charge and spin may undergo an exchange—as U passes through U_c, the scaling of the charge compressibility switches from metallic behavior (power-law enhancement) to insulating behavior (approaching a constant), while the scaling of the spin gap remains critical ($\sim 1/L$). This "scaling exchange" is a phenomenon not explicitly described in standard Luttinger liquid theory—which typically treats the charge and spin sectors separately, without describing the scaling transfer relationship between them.

Our conjecture is: for a given finite L, there exists a $U_c(L)$ such that for $U < U_c$, the charge compressibility exhibits metallic scaling ($\chi \sim L^\alpha$, $\alpha > 0$), and for $U > U_c$, χ approaches a constant. $U_c(L)$ itself should approach zero with L as $U_c(L) \sim 1/L$ (consistent with the Lieb-Wu thermodynamic limit result), but its prefactor may be related to the topological index C. Systematic testing of this conjecture can be performed through DMRG calculations of charge compressibility and spin gap at different U and L.

Prediction 3: Cross-sectoral discord transfer—when the charge gap suppresses scaling, the discord does not disappear. In the Hubbard model, the charge compressibility shows no critical scaling (in the charge sector, the discord is "silenced" by the Mott gap), but the spin gap shows clear critical scaling (in the spin sector, the discord "manifests" in the form of gapless excitations). These are not two independent physical facts—they are different manifestations of the same deep physical mechanism (Mott physics) in two degrees of freedom.

Our conjecture is: near the Mott transition point (i.e., in the region $U \approx U_c(L)$), there exists a cross-sectoral "discord transfer" mechanism. When the discord density in the charge sector (which can be quantitatively characterized by the derivative of compressibility with respect to L) drops from positive values in the metallic region to zero in the insulating region, the discord density in the spin sector (which can be quantitatively characterized by the absolute value of the derivative of the spin gap with respect to L) remains constant. This means that discord is not eliminated but transferred from one sector to another—discord is the driving force of system evolution, and harmony is not the elimination of discord but the dynamic equilibrium of discord.

The testing scheme for this conjecture is: at multiple U/t values (from 0 to 4) and multiple L values, simultaneously compute the charge compressibility χ and the spin gap Δ_s, and extract their scaling exponents with respect to L, $\alpha_{\text{charge}}(U)$ and $\alpha_{\text{spin}}(U)$. If the cross-sectoral transfer mechanism holds, then there should exist a clear complementary relationship between $\alpha_{\text{charge}}(U)$ and $\alpha_{\text{spin}}(U)$: $\alpha_{\text{charge}}(U) + \alpha_{\text{spin}}(U) \approx \text{const}$, across the transition region where U passes through $U_c(L)$.

If this prediction is supported by data, it will be a significant piece of evidence that the EDRN framework transcends standard strongly correlated physics—standard theory does not contain such a cross-sectoral scaling exponent conservation law.

F. Limitations of This Work

We honestly declare the known limitations of this work:

(1) Limitation of system size: Constrained by local computational resources (pure Python mode, $\chi_{\max} = 100$), our maximum chain lengths are $L = 200$ (charge compressibility) and $L = 100$ (spin gap and Prediction 1). For calculations at larger L (e.g., $L = 500$ or beyond), larger bond dimensions ($\chi_{\max} \geq 500$) and more efficient computational environments (e.g., GPU acceleration or Cython compilation) may be necessary. The finite-size scaling analysis of this paper is based on currently accessible system sizes; the validity of its extrapolation to the thermodynamic limit needs to be verified through larger-scale calculations.

(2) Limitation of parameter range: All calculations in this paper are fixed at $U/t = 4$ and half-filling. Testing Prediction 2 and Prediction 3 requires DMRG scans across a wider parameter space of U/t and filling factors. Whether the linear coefficients a and b in Li's Scaling are universal also needs to be tested at different U/t values. This is the focus of subsequent work.

(3) Single methodological path: Our calculations rely entirely on DMRG as the sole numerical method. Certain conclusions among the predictions (e.g., the $1/L$ scaling of the spin gap, Li's Scaling) can in principle be cross-validated using other methods (such as quantum Monte Carlo, exact diagonalization, cold atom quantum simulation). Multi-method cross-validation will further enhance the reliability of the conclusions.

(4) Causality not yet confirmed: Li's Scaling currently manifests as an empirical regularity—five data points show a strong linear correlation. However, whether $A(C)$ is determined by C (causality), or whether the two merely drift in the same direction by coincidence at finite sizes (correlation), has not been rigorously proven. To confirm causality, the experiment needs to be repeated at different U/t values to test whether the linear relationship still holds.

(5) Translation requirement for theoretical language: The EDRN framework employs a non-standard conceptual system (relational network, discord dynamics, silent discord, etc.). The correspondence between these concepts and the standard language of condensed matter physics (Hamiltonian, correlation function, scaling law) has been established as clearly as possible in this paper, but completely eliminating the "translation error" between the two languages requires longer-term dialogue and accumulation.

These boundary declarations are not a failure of the theoretical framework, but evidence of its honesty.

VIII. CONCLUSION

This paper completes a full scientific cycle from theoretical construction to numerical verification, reporting the first independent numerical verification of the EDRN framework. The core findings are as follows:

1. **Numerical finding 1 (Charge sector):** Through the two-sector DMRG method, calculations on the 1D half-filled Hubbard model ($U/t = 4$) for $L = 20$ to 200 reveal that the charge gap approaches a constant $\Delta E \approx 1.35t$, the charge compressibility approaches a constant $\chi \approx 0.74$, and the fitted scaling exponent is $\alpha \approx -0.03$. The original theoretical prediction $\alpha = 1.75$ fails in the charge sector. The system is in the Mott insulating phase, with the charge degree of freedom frozen by a gap.

2. **Theory correction:** Based on experimental evidence, we analyze that the failure of the prediction lies not in the theoretical framework itself, but in the critical regime being mislocated to the charge sector. The anomalous dimension $\eta = 1/4$ predicted by the EDRN framework, and its scaling consequences, must point to the spin sector—where gapless critical behavior is maintained through Heisenberg exchange interaction. A corrected prediction is proposed: spin gap $\Delta_s \sim 1/L$.

3. **Numerical finding 2 (Spin sector):** DMRG calculations of the spin gap for the same system over $L = 20$ to 100 show Δ_s approaching zero as $\sim 1/L$, with a fitted scaling exponent $\alpha = -1.05 \pm 0.03$, $R^2 = 0.996$. The corrected prediction is supported by data. The spin sector is the true stage for the EDRN critical scaling behavior.

4. **Numerical finding 3 (Li's Scaling):** A direct DMRG test of Prediction 1 is performed. Five independent data points ($L = 20$ to 100) show a

clear linear correlation between the spin gap prefactor $A(C) = \Delta_s \times L$ and the topological connectivity index C ($R^2 = 0.998$), establishing $A(C) = (12.1 \pm 0.3) \times C - (8.5 \pm 0.3)$. We formally name this newly discovered empirical scaling law **Li's Scaling**. This is a structure-response correspondence not previously described in standard Luttinger liquid theory—standard theory treats the prefactor as a non-universal constant, while the EDRN framework predicts its connection to the network's topological characteristics.

5. **Methodological demonstration:** This study demonstrates how to guard against the instrumental rationality paradox in numerical computation—through self-correction from the three-sector to the two-sector method, honestly facing negative data, and correcting theoretical boundaries based on physical intuition rather than forcibly "repairing" data.

6. **Two new predictions:** Based on the established theoretical foundation and numerical methods, we propose two predictions for further testing: (1) the exchange of charge and spin scaling behaviors at an effective critical U value in finite sizes, and (2) cross-sectoral discord transfer near the Mott transition point.

The fundamental contribution of this work does not lie in verifying a specific numerical value (1.75), but in completing a full scientific cycle of prediction-test-correction-discovery. The EDRN framework, starting from a relational network perspective, makes testable predictions, accepts the verdict of data, corrects itself based on feedback, and in the process discovers a new scaling relation—Li's Scaling—not previously described by standard theory. The relational network perspective is not a static declaration, but a scientific program capable of making testable predictions, accepting data verdicts, correcting itself based on feedback, and producing new discoveries. Three DMRG datasets—the eight-point dataset of charge compressibility, the five-point dataset of the spin gap, and the five-point dataset of Prediction 1—together constitute the first empirical foundation of this program.

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